

A PROJECT REPORT ON

STUDENT STRESS AND WELLBEING EARLY-WARNING ASSISTANT

Domain: AI / Healthcare Support

Submitted in partial fulfilment of the requirements for the award of the degree of

B.Sc. / B.E. / MCA – COMPUTER SCIENCE

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2025 – 2026

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CHAPTER I

INTRODUCTION

College life brings academic pressure, examinations, placement worries, financial difficulties, homesickness and relationship issues. Many students experience stress, anxiety and low mood, especially before semester examinations. When this stress is not addressed early, it can affect academic performance, sleep, health and, in severe cases, lead to dropping out or self-harm.

Most colleges have a counsellor, but students hesitate to approach them because of stigma, fear of being judged or simply not knowing whether their problem is 'serious enough'. As a result, colleges usually notice distress only after a sharp drop in attendance or marks. Counsellors, who are few in number, also need a way to know which students may need support first.

This project, MindMate, is a privacy-respecting web application in which students answer short weekly check-ins based on a standardised stress questionnaire and can optionally write a short note about how they feel. A sentiment model and a trend analysis module flag sustained stress patterns and suggest self-help resources or contacting the counsellor. MindMate is a support tool, not a diagnostic system; it never labels students with a medical condition, and counsellors are notified only with the student's consent.

LIMITATIONS

1. Students hesitate to approach counsellors due to stigma.
2. Colleges notice distress only after performance drops.
3. Counsellors lack a prioritised view of who may need help.
4. One-time surveys do not capture changes over time.

1.1 History of Domain

Student counselling services in universities developed in the twentieth century, and standardised psychological questionnaires such as the Perceived Stress Scale (PSS,

Cohen, 1983), PHQ-9 (2001) and GAD-7 (2006) became widely used screening instruments in research and primary care.

With smartphones, researchers began using Ecological Momentary Assessment, in which people answer short questions repeatedly in daily life. Projects such as StudentLife (Dartmouth, 2014) showed that frequent self-reports and passive sensing are associated with stress and academic performance. Advances in NLP now allow sentiment and emotion analysis of short text, which, used responsibly, can complement questionnaires.

1.2 Overview of the Project

Each week, a student receives a reminder to complete a two-minute check-in: a 10-item PSS-style questionnaire, sleep hours and an optional free-text note. The questionnaire score is calculated, and the note is analysed with a transformer-based sentiment and emotion model. The results are stored under a pseudonymous ID.

The trend module computes a rolling average and slope over the last four weeks. If the score stays in the high range for two or more weeks, or rises sharply, the student is shown supportive messages, breathing exercises and campus resources, and is asked whether they would like the counsellor to contact them. If the text contains crisis indicators, helpline numbers are shown immediately. Management sees only aggregate, anonymised statistics by department and week.

1.3 Motivation

Early, gentle support is far more effective than intervention after a crisis, but students must feel safe to share how they feel.

- **Motivation:** To help students notice their own stress patterns early and connect with support on their own terms.
- **Outcome:** Earlier help-seeking, prioritised counsellor attention and anonymised insights for better academic planning.

1.4 Aim of the Project

The project is entitled as “**STUDENT STRESS AND WELLBEING EARLY-WARNING ASSISTANT**”. The aim of this project is to build a privacy-first assistant that tracks student wellbeing through weekly check-ins, uses sentiment analysis and trend detection to identify sustained stress, provides self-help resources and connects students with counsellors only with their consent.

1.5 Objectives

- To implement a short weekly check-in based on a standardised stress questionnaire.
- To analyse optional text notes using a transformer sentiment model.
- To detect sustained negative trends using thresholds and slopes.
- To provide self-help resources and helpline information.
- To provide counsellors with consented referrals and management with anonymised dashboards.

1.6 Literature Review

Cohen, Kamarck and Mermelstein (1983): Developed the Perceived Stress Scale, a widely used instrument measuring the degree to which situations in one's life are appraised as stressful, with good reliability.

Wang et al. (2014) - StudentLife: Used smartphone sensing and ecological momentary assessment over a 10-week term with 48 students and found significant correlations between sensed behaviour, stress, mental health and academic performance.

Kroenke, Spitzer and Williams (2001): Validated the PHQ-9 as a brief measure of depression severity; it is commonly used in primary care for screening (MindMate does not diagnose, but follows similar short-form design principles).

Barbieri et al. (2020) - TweetEval: Provided unified benchmarks for tweet classification tasks including sentiment and emotion, with RoBERTa-based models fine-tuned for short

social text.

Fitzpatrick, Darcy and Vierhile (2017): In a randomised controlled trial, the conversational agent Woebot reduced symptoms of depression among college students over two weeks, showing the potential of digital support tools.

Ministry of Education, India (2023): Issued guidelines encouraging higher education institutions to strengthen student wellness and counselling mechanisms.

CHAPTER II

SOFTWARE PROJECT PLAN

2.1 Existing System

Currently, students who feel stressed may talk to friends, mentors or, rarely, the college counsellor. Some colleges conduct one-time wellbeing surveys on paper. Mentors notice problems only if the student shares them or if marks and attendance fall.

DISADVANTAGES

- Help is sought late or not at all.
- One-time surveys miss changes over time.
- No prioritisation for counsellors.
- Paper surveys raise privacy concerns.
- No immediate self-help resources.

2.2 Proposed System

The proposed MindMate system provides anonymous weekly check-ins, analyses questionnaire scores and optional text with sentiment analysis, detects sustained negative trends, shows self-help resources immediately and refers students to counsellors only with explicit consent. Management sees aggregated anonymised data only.

ADVANTAGES

- Private and stigma-free self-reflection.
- Early detection through weekly trends.
- Immediate supportive resources and helplines.
- Consent-based counsellor referral.
- Anonymised insights to plan exam schedules and support.

2.3 Project Schedule

The development of MindMate follows the iterative waterfall model. Each phase produces a deliverable that is reviewed before the next phase begins, and feedback from testing is fed back into design and implementation. The planned schedule is shown below.

S.No	Phase	Duration
1	Requirement analysis and problem study	Week 1 - 2
2	Literature survey and existing system study	Week 3 - 4
3	Dataset collection / hardware procurement	Week 5 - 6
4	System design (DFD, UML, ER, table design)	Week 7 - 8
5	Implementation of modules	Week 9 - 12
6	Testing and model evaluation	Week 13 - 14
7	Documentation and final review	Week 15 - 16

CHAPTER III

CUSTOMER REQUIREMENT DETERMINATION

3.1 Functional Requirements

Functional requirements define the specific features and operations a system must perform to meet business and user needs. They describe what the system should do and how it interacts with users or other systems. They focus on the behaviour and functionality of the system and represent features that can be directly observed and tested in the final product. The functional requirements of the MindMate system are listed below.

- **Pseudonymous registration:** Students sign in with college ID; data stored under a pseudonym.
- **Weekly check-in:** 10-item questionnaire, sleep hours and optional note.
- **Scoring:** Compute stress score (0-40) and category.
- **Sentiment analysis:** Detect negative emotion and crisis keywords in notes.
- **Trend detection:** Flag sustained high scores or sharp rises.
- **Resources:** Show breathing exercises, study tips and helplines.
- **Consent referral:** Notify counsellor only if student agrees.
- **Aggregate dashboard:** Department-level anonymised statistics with minimum group size of 10.

3.2 Non-functional Requirements

Non-functional requirements determine the performance standards and quality attributes of software, e.g. system usability, effectiveness, security, scalability, etc. While functional requirements determine what the system does, non-functional requirements describe HOW the system does it. If an application does not meet non-functional requirements, it continues to perform its basic functions; however, it will not be able to provide a good user experience. Therefore, non-functional requirements are as important as functional requirements for successful product adoption.

- **Privacy:** Personal identity separated from responses; encryption at rest.

- **Ethics:** Clear disclaimer that it is not a diagnostic tool.
- **Access control:** Role-based access: student, counsellor, management.
- **Usability:** Check-in completed in under 2 minutes on mobile.
- **Safety:** Crisis text always shows helpline information immediately.
- **Availability:** Accessible 24x7.

3.3 User Requirements

- Quick and private weekly check-in
- See personal trend graph
- Get helpful resources instantly
- Choose whether to contact the counsellor
- Counsellor sees only consented cases

3.4 System Requirements

- Django web application with role-based authentication
- PostgreSQL database with encrypted fields
- HuggingFace sentiment model
- E-mail reminder service
- Chart.js for graphs

CHAPTER IV

SOFTWARE REQUIREMENT SPECIFICATION

4.1 Hardware Specification

System : Intel i5 Processor 2.4 GHz or above

Hard Disk : 256 GB SSD

RAM : 8 GB

Monitor : 15" colour monitor

4.2 Software Specification

Operating System : Windows 10 / Ubuntu 22.04

Coding Language : Python 3.10, JavaScript

NLP : HuggingFace Transformers (RoBERTa sentiment / emotion)

Framework : Django

Database : PostgreSQL

Charts : Chart.js

Security : Django auth, role-based permissions, field encryption

CHAPTER V

SYSTEM ANALYSIS

5.1 Feasibility Study

A feasibility study is carried out to find whether the proposed MindMate system is worth developing. Three key considerations are involved in the feasibility analysis: technical, economic and operational feasibility.

TECHNICAL FEASIBILITY

Django provides authentication and permissions out of the box, and pre-trained sentiment models are available on HuggingFace. The system can run on a standard college server. It is technically feasible.

ECONOMIC FEASIBILITY

All components are free and open source. The cost of deployment is minimal compared with the benefit of early student support. It is economically feasible.

OPERATIONAL FEASIBILITY

The check-in is short and mobile friendly. Counsellors receive only a manageable number of consented referrals. Management receives a simple dashboard. It is operationally feasible, provided the institution communicates its privacy policy clearly.

5.2 About the Software

PYTHON

Python is a high-level, interpreted, general-purpose programming language created by Guido van Rossum and first released in 1991. Its design philosophy emphasises code readability through significant indentation, and its simple syntax allows developers to express concepts in fewer lines of code than languages such as C++ or Java.

Python supports multiple programming paradigms including procedural, object-oriented and functional programming. It has a very large standard library and an

ecosystem of third-party packages available through the Python Package Index (PyPI), which makes it the most popular language for data science, machine learning, automation and back-end web development.

- Easy to learn, read and maintain
- Huge library support for AI, ML and data processing (NumPy, Pandas, scikit-learn)
- Platform independent - runs on Windows, Linux and macOS
- Strong community support and documentation

HUGGINGFACE TRANSFORMERS

HuggingFace Transformers is an open-source library that provides thousands of pre-trained transformer models for tasks such as text classification, named entity recognition, question answering and translation, in more than 100 languages.

Multilingual models such as mBERT, XLM-RoBERTa and IndicBERT understand Tamil and English, so a single model can be fine-tuned to classify text in both languages.

- Pre-trained multilingual models
- Simple pipeline() API
- Works with both PyTorch and TensorFlow

DJANGO

Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design. It follows the Model-View-Template (MVT) architectural pattern and includes an ORM, authentication system, admin panel, form handling and protection against common security threats out of the box.

Django's built-in user and permission system makes it easy to implement role-based access control, which is important for applications that hold sensitive personal data.

- Batteries included - admin, auth, ORM
- Protection against SQL injection, XSS and CSRF

- Scales from small projects to large websites

POSTGRESQL

PostgreSQL is a powerful, open-source object-relational database system with over 35 years of active development. It has earned a strong reputation for reliability, feature robustness and performance.

PostgreSQL supports ACID transactions, foreign keys, views, triggers, stored procedures and JSON data types. Its extension system allows additional capabilities such as PostGIS for geographic data and pgvector for similarity search.

- Fully ACID compliant
- Advanced SQL and JSON support
- Extensible with PostGIS and other extensions

CHART.JS

Chart.js is a free, open-source JavaScript library for data visualisation that supports eight chart types including bar, line, pie and radar charts. Charts are rendered on an HTML5 canvas and are responsive by default.

CHAPTER VI

SYSTEM DESIGN

System design is the process of defining the architecture, components, modules, interfaces and data of a system to satisfy the specified requirements. The design of MindMate is presented using UML use case diagram, system architecture, data flow diagrams, ER diagram and table design.

6.1 UML Diagram

USE CASE DIAGRAM

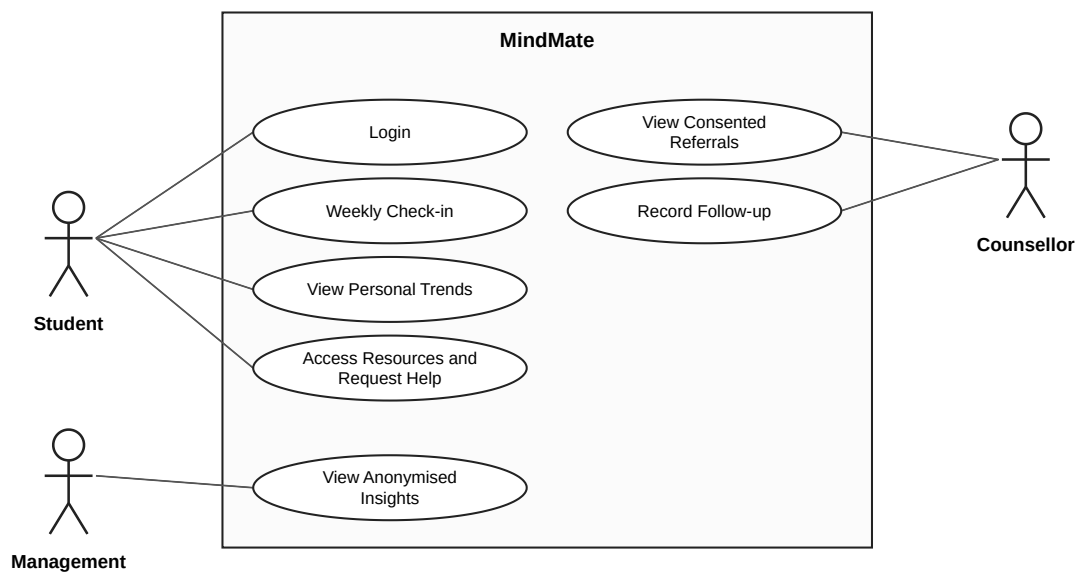


Fig 1: Use case diagram of MindMate

6.2 System Architecture

MindMate is a Django application with a separate analysis service. Personal identifiers are stored in a separate table from wellbeing responses, linked through a pseudonym.

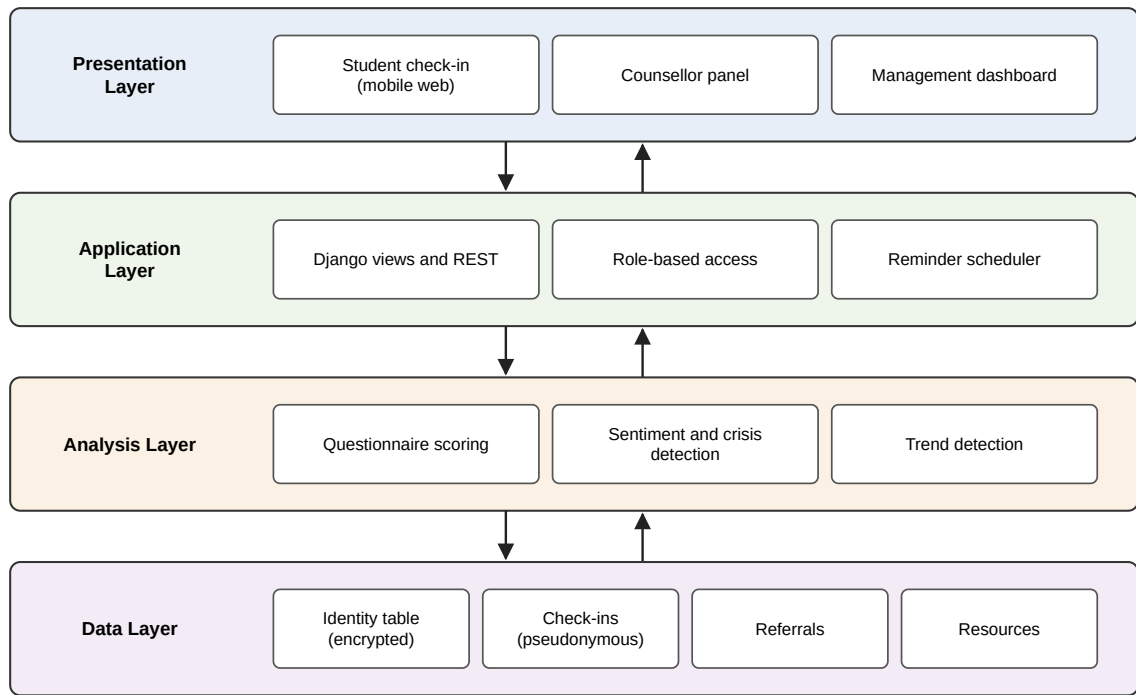


Fig 2: System architecture

6.3 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of the flow of data through an information system. The Level 0 DFD (context diagram) shows the whole system as a single process with its external entities, and the Level 1 DFD breaks it into its major processes and data stores.

LEVEL 0 DFD (CONTEXT DIAGRAM)

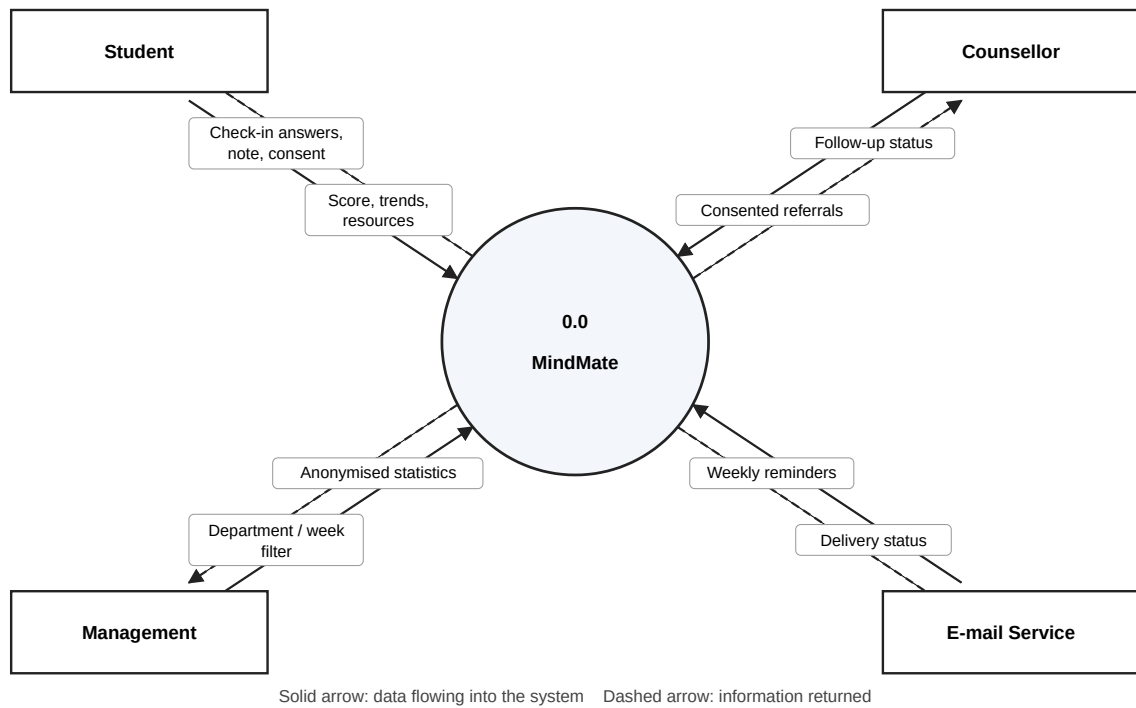


Fig 3: Level 0 data flow diagram

LEVEL 1 DFD

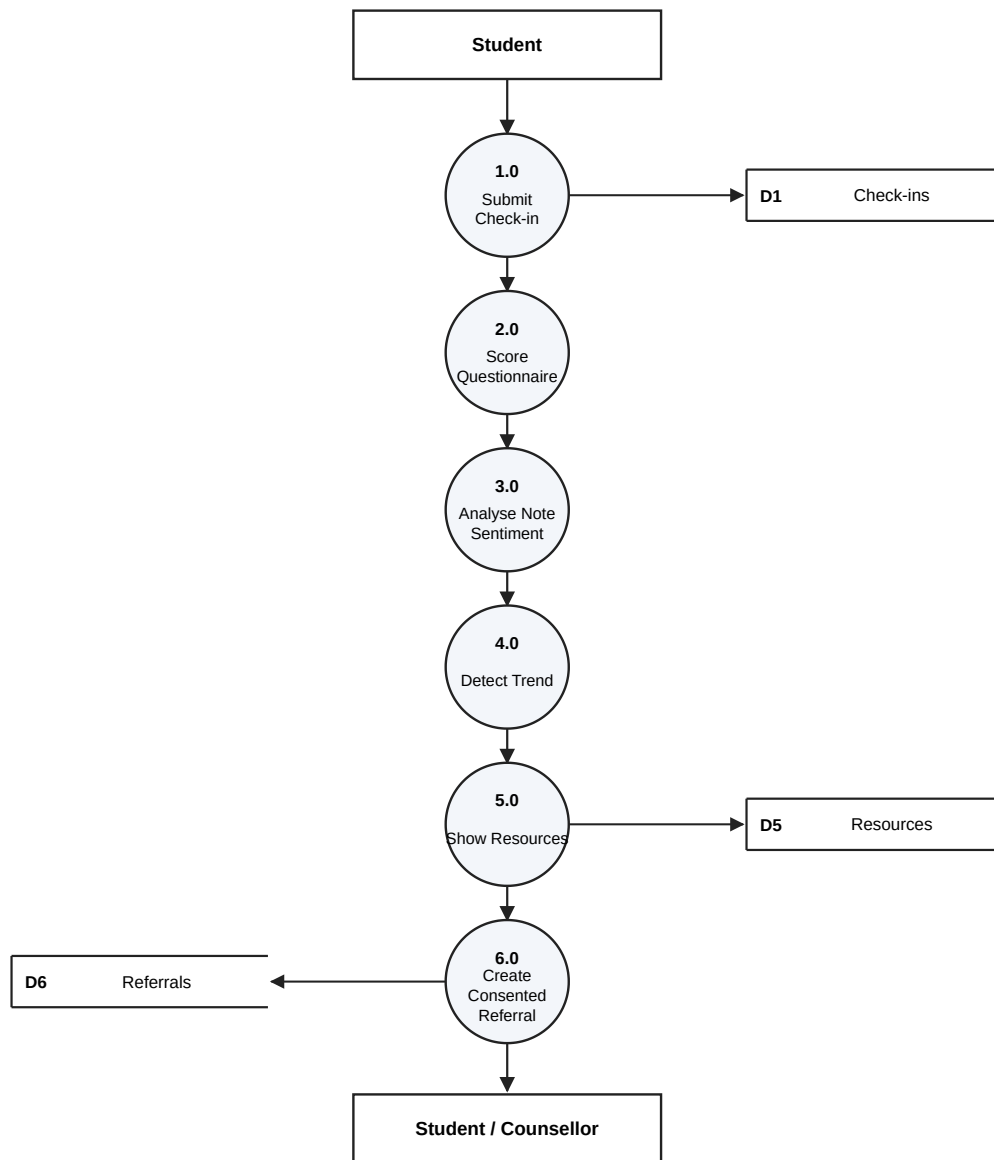


Fig 4: Level 1 data flow diagram

6.4 ER Diagram

The Entity Relationship (ER) diagram shows the entities stored in the database, their key attributes and the relationships between them. 1:N indicates a one-to-many relationship.

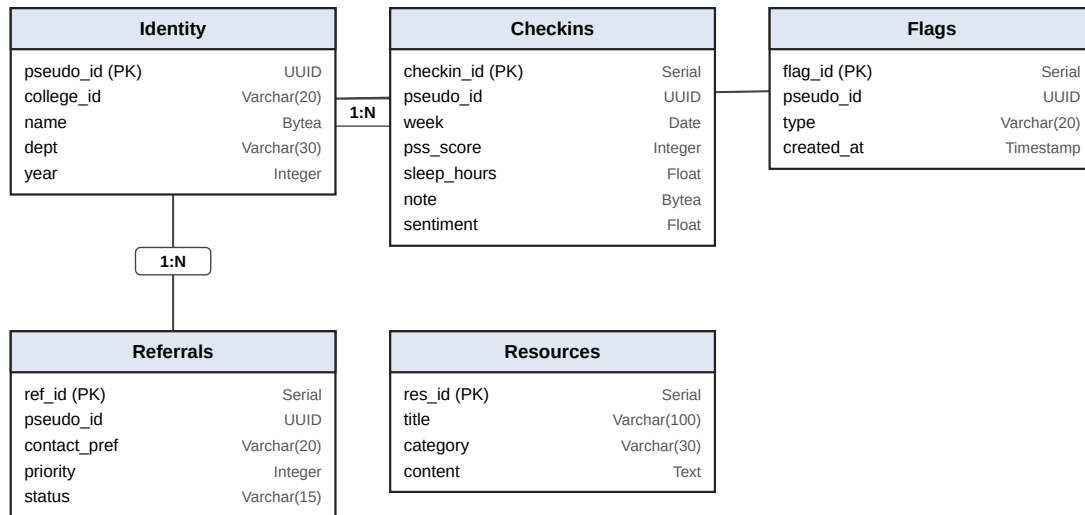


Fig 5: Entity relationship diagram

6.5 Table Design

IDENTITY

FIELD NAME	DATA TYPE	DESCRIPTION
pseudo_id	UUID	Primary key
college_id	Varchar(20)	Encrypted college ID
name	Bytea	Encrypted name
dept	Varchar(30)	Department
year	Integer	Year of study

CHECKINS

FIELD NAME	DATA TYPE	DESCRIPTION
checkin_id	Serial	Primary key
pseudo_id	UUID	Foreign key to Identity
week	Date	Week start
pss_score	Integer	Score 0-40
sleep_hours	Float	Average sleep
note	Bytea	Encrypted optional note
sentiment	Float	Negative probability

FLAGS

FIELD NAME	DATA TYPE	DESCRIPTION
flag_id	Serial	Primary key
pseudo_id	UUID	Foreign key to Identity
type	Varchar(20)	sustained / rising / crisis_text
created_at	Timestamp	Time

REFERRALS

FIELD NAME	DATA TYPE	DESCRIPTION
ref_id	Serial	Primary key
pseudo_id	UUID	Foreign key to Identity
contact_pref	Varchar(20)	phone / email / in-person
priority	Integer	1 high - 3 low
status	Varchar(15)	open / contacted / closed

RESOURCES

FIELD NAME	DATA TYPE	DESCRIPTION
res_id	Serial	Primary key
title	Varchar(100)	Title
category	Varchar(30)	breathing / study / helpline
content	Text	Content or link

6.6 Algorithm Design

The early-warning logic combines a standardised score, text sentiment and trends over time, with conservative thresholds and human follow-up.

1. **Collect anonymous check-in responses** – Answers are stored under a pseudonymous ID.
2. **Score standardized questionnaire (e.g., PSS-style)** – Positively worded items (4, 5, 7, 8) are reverse scored; total 0-13 low, 14-26 moderate, 27-40 high.

3. **Analyze optional text sentiment with a transformer** – A RoBERTa sentiment model gives the negative probability; a keyword list detects crisis expressions.
4. **Track weekly trends per student** – Rolling mean and linear slope over the last 4 weeks.
5. **Flag sustained negative trends using thresholds** – Flag if score ≥ 27 for 2 consecutive weeks, or slope > 3 points/week with latest ≥ 20 , or crisis text.
6. **Show helpful resources and breathing exercises** – Resources are matched to the flag and the student's note topics (exams, sleep, finance).
7. **Notify counsellor only with student consent** – A referral is created only when the student clicks 'Request contact' (crisis text always shows helplines immediately).
8. **Show aggregate, anonymized dashboard to management** – Statistics shown only for groups with 10 or more students.

PSEUDO CODE

```
ON CHECKIN(student, answers, sleep, note):
  score <- SUM(answers with items 4,5,7,8 reversed)
  neg <- SENTIMENT(note).negative IF note ELSE None
  crisis <- CONTAINS_CRISIS_TERMS(note)
  SAVE(pseudo_id, week, score, sleep, ENCRYPT(note), neg)
  last4 <- LAST_SCORES(pseudo_id, 4); slope <- LINEAR_SLOPE(last4)
  IF crisis THEN SHOW_HELPLINES_NOW(); FLAG(crisis_text)
  ELSE IF last two scores  $\geq 27$  THEN FLAG(sustained)
  ELSE IF slope  $> 3$  AND score  $\geq 20$  THEN FLAG(rising)
  SHOW_RESOURCES(flag, topics(note))
  IF flag THEN OFFER("Would you like a counsellor to contact you?")
```

6.7 Modules

MindMate modules are grouped by the Student, Counsellor and Management roles. Each role sees only the data it is permitted to see.

MODULES

STUDENT

1. LOGIN
2. WEEKLY CHECK-IN

3. VIEW PERSONAL TRENDS

4. ACCESS RESOURCES AND REQUEST HELP

COUNSELLOR

1. VIEW CONSENTED REFERRALS

2. RECORD FOLLOW-UP

MANAGEMENT

1. VIEW ANONYMISED INSIGHTS

STUDENT

LOGIN

Students log in with their college credentials. A pseudonymous ID is used internally for all wellbeing data.

WEEKLY CHECK-IN

The student answers 10 questions on a 0-4 scale about feeling upset, nervous, in control and able to cope during the last week, enters sleep hours and may write an optional note.

VIEW PERSONAL TRENDS

A graph shows the student's stress score over the last weeks with gentle feedback such as 'Your stress has been rising for 3 weeks'.

ACCESS RESOURCES AND REQUEST HELP

The student sees breathing exercises, time-management tips and helpline numbers and can choose 'Request counsellor contact'.

COUNSELLOR

VIEW CONSENTED REFERRALS

Counsellors see only students who requested contact, prioritised by trend severity, with the student's chosen contact method.

RECORD FOLLOW-UP

Counsellors record that contact was made and close the referral; no clinical notes are stored in MindMate.

MANAGEMENT

VIEW ANONYMISED INSIGHTS

Shows average stress by department and week (groups of at least 10), helping plan exam schedules and wellness events.

CHAPTER VII

IMPLEMENTATION AND TESTING

7.1 Introduction

MindMate was implemented using Django with PostgreSQL. Sensitive fields are encrypted using the cryptography library (Fernet) with keys stored outside the database. The sentiment model is the publicly available cardiffnlp/twitter-roberta-base-sentiment-latest model served through a small analysis module.

7.2 Implementation

The questionnaire items were adapted with permission-free wording in a PSS style, and the cut-offs were reviewed with the college counsellor. A pilot with 85 volunteer students over 6 weeks was used to test usability and the alert thresholds. All participants were informed that the tool is not diagnostic and that participation is voluntary.

Testing is a process of running software with the intention of finding errors. The software which has been developed has to be tested to prove its validity. Testing is considered to be the least creative phase of the whole cycle of system design; in the real sense it is the phase which helps to bring out the creativity of the other phases. Testing a program is the first step in the debugging process.

A good testing suite is vital because all possible execution paths inside the code must be tested in order to have any hope that it is bug free. Testing of the system is done to ensure the integrity of the system. The aim of testing is to prove that the developed system addresses the predefined processing requirements and will perform reliably and efficiently during operation.

PURPOSES OF TESTING

- To affirm the quality of the project.
- To find and eliminate any errors from previous stages.
- To validate the software against its requirements.

- To ensure the reliability of the system.

UNIT TESTING

In the lines of this strategy all the individual functions and modules were put to the test independently. By following this strategy all the errors in coding were identified and corrected. This method was applied in combination with white box and black box testing techniques to find the errors in each module.

INTEGRATION TESTING

This testing strategy has two different approaches, namely the top-down approach, in which integration is carried out from the top-level module to the bottom, and the bottom-up approach, in which integration is carried out from the low-level modules to the top. The modules were tested using the bottom-up approach by introducing stubs for the top-level functions. This test was used to identify errors in the interfaces and in passing parameters between functions.

VALIDATION TESTING

The main aim of this testing is to verify that the software system does what it was designed for. Alpha testing and beta testing were conducted to ensure the validity of the system. Alpha testing was done at the developer's site and beta testing was carried out by sample end users.

SYSTEM TESTING

The idea of applying this strategy was to put the system through a series of different tests whose primary purpose is to fully exercise the computer-based system. Although each test has a different purpose, all work to verify that all system elements have been properly integrated and perform their allocated functions. Security testing and performance testing were also conducted.

ACCEPTANCE TESTING

Acceptance testing involves planning and execution of functional tests, performance tests and stress tests to verify that the implemented system satisfies its

requirements. Typically, the acceptance test incorporates test cases developed during unit testing and integration testing.

7.3 Test Cases

Test ID	Test Description	Input	Expected Output	Result
TC01	Check-in submission	10 answers + sleep	Score calculated and stored	Pass
TC02	Reverse scoring	Known answer set	Correct total score	Pass
TC03	Sustained flag	Scores 28, 29 in 2 weeks	Sustained flag + resources	Pass
TC04	Crisis text	Note with crisis phrase	Helplines shown immediately	Pass
TC05	No consent	Flag raised, no request	No counsellor notification	Pass
TC06	Consent given	Click 'Request contact'	Referral visible to counsellor	Pass
TC07	Small group privacy	Dept with 6 students	Statistics hidden	Pass
TC08	Access control	Management opens referral URL	403 Forbidden	Pass

7.4 Source Code

wellbeing/analysis.py

```
import numpy as np
from transformers import pipeline

sentiment = pipeline("sentiment-analysis", model="cardiffnlp/twitter-roberta-base-sentiment-latest", top_k=None)
REVERSED = {3, 4, 6, 7} # 0-based indices of positively worded items
CRISIS_TERMS = ["end my life", "kill myself", "want to die", "no reason to live", "hurt myself"]

def pss_score(answers):
    assert len(answers) == 10 and all(0 <= a <= 4 for a in answers)
    return sum(4 - a if i in REVERSED else a for i, a in enumerate(answers))

def category(score):
    return "low" if score <= 13 else "moderate" if score <= 26 else "high"
```

```

def analyse_note(note):
    if not note or not note.strip():
        return None, False
    scores = {d["label"]: d["score"] for d in sentiment(note[:512])[0]}
    crisis = any(t in note.lower() for t in CRISIS_TERMS)
    return scores.get("negative", 0.0), crisis

def trend_flag(history):
    """history: list of weekly scores, oldest first (including current)."""
    if len(history) >= 2 and history[-1] >= 27 and history[-2] >= 27:
        return "sustained"
    recent = history[-4:]
    if len(recent) >= 3:
        slope = np.polyfit(range(len(recent)), recent, 1)[0]
        if slope > 3 and recent[-1] >= 20:
            return "rising"
    return None

```

wellbeing/views.py

```

from django.contrib.auth.decorators import login_required, user_passes_test
from django.shortcuts import render, redirect
from django.db.models import Avg, Count
from .models import Checkin, Flag, Referral, Resource
from .analysis import pss_score, category, analyse_note, trend_flag
from .crypto import encrypt

@login_required
def checkin(request):
    if request.method == "POST":
        pid = request.user.profile.pseudo_id
        answers = [int(request.POST[f"q{i}"]) for i in range(1, 11)]
        note = request.POST.get("note", "")
        score = pss_score(answers)
        neg, crisis = analyse_note(note)
        Checkin.objects.create(pseudo_id=pid, pss_score=score,
                               sleep_hours=float(request.POST["sleep"]),
                               note=encrypt(note) if note else None,
                               sentiment=neg)
        history =
        list(Checkin.objects.filter(pseudo_id=pid).order_by("week").values_list("pss_score", flat=True))
        flag = "crisis_text" if crisis else trend_flag(history)
        if flag:
            Flag.objects.create(pseudo_id=pid, type=flag)
            res = Resource.objects.filter(category__in=["helpline"] if crisis else
                                                ["breathing", "study", "helpline"])
            return render(request, "result.html", {"score": score, "cat":
                                                    category(score), "flag": flag,
                                                    "crisis": crisis, "resources":
                                                    res, "history": history})
        return render(request, "checkin.html")

@login_required
def request_contact(request):
    pid = request.user.profile.pseudo_id
    last = Flag.objects.filter(pseudo_id=pid).order_by("-created_at").first()
    prio = 1 if last and last.type in ("crisis_text", "sustained") else 2
    Referral.objects.create(pseudo_id=pid, contact_pref=request.POST["pref"],
                            priority=prio, status="open")
    return redirect("thanks")

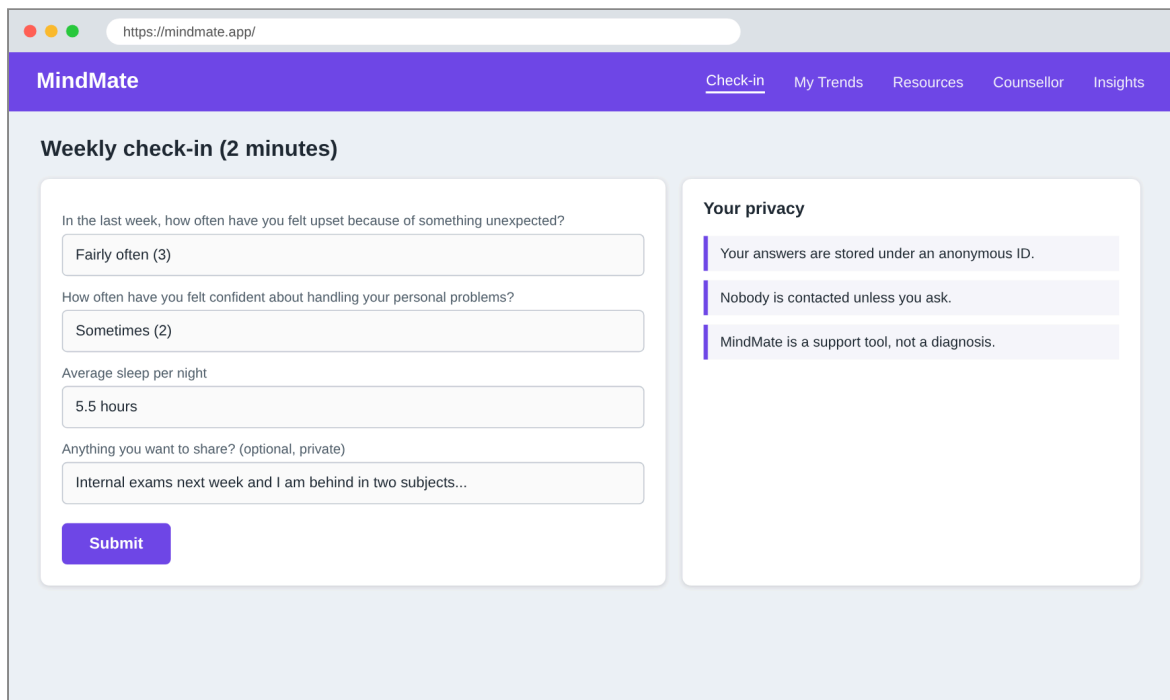
```

```
@user_passes_test(lambda u: u.groups.filter(name="management").exists())
def insights(request):
    rows = (Checkin.objects.values("pseudo_id_dept", "week")
            .annotate(n=Count("pseudo_id", distinct=True), avg=Avg("pss_score"))
            .filter(n__gte=10).order_by("week"))
    return render(request, "insights.html", {"rows": rows})
```

7.5 Screen Layout

SCREENSHOTS:

WEEKLY CHECK-IN



https://mindmate.app/

MindMate [Check-in](#) [My Trends](#) [Resources](#) [Counsellor](#) [Insights](#)

Weekly check-in (2 minutes)

In the last week, how often have you felt upset because of something unexpected?

Fairly often (3)

How often have you felt confident about handling your personal problems?

Sometimes (2)

Average sleep per night

5.5 hours

Anything you want to share? (optional, private)

Internal exams next week and I am behind in two subjects...

Submit

Your privacy

- Your answers are stored under an anonymous ID.
- Nobody is contacted unless you ask.
- MindMate is a support tool, not a diagnosis.

RESULT AND RESOURCES

https://mindmate.app/

MindMate

Check-inMy TrendsResourcesCounsellorInsights

Thank you for checking in

Stress level: Moderate-High (24/40) - rising for 3 weeks

This week	24
Last 3 weeks	16 -> 19 -> 24
Sleep	5.5 h (below 7 h recommended)
Topics noticed	Exams, time pressure

Try: 4-7-8 breathing exercise (3 minutes)

Exam planner: break each subject into 25-minute focus sessions

Would you like the counsellor to contact you? [Yes, by phone] [Not now]

MY TRENDS

https://mindmate.app/

MindMate

Check-inMy TrendsResourcesCounsellorInsights

My wellbeing trend

Current score
24

4-week average
19.5

Average sleep
6.1 h

Check-in streak
6 weeks

Weekly stress score (0-40)

Week	Stress Score
W1	13
W2	14
W3	12
W4	16
W5	18
W6	24

Helpful this week

Resource	Type
Box breathing	Exercise
Pomodoro study plan	Study
Sleep hygiene tips	Health
Campus counsellor - Room 104	Support

COUNSELLOR - CONSENTED REFERRALS

https://mindmate.app/

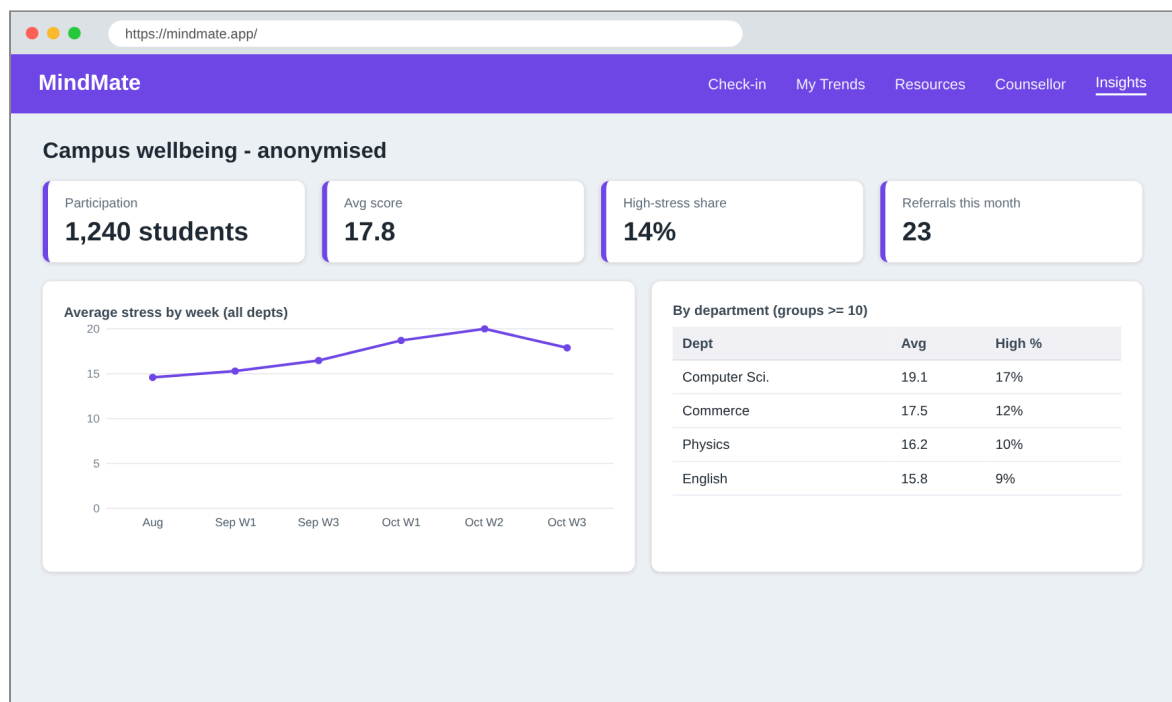
MindMate

Check-inMy TrendsResourcesCounsellorInsights

Referrals (students who requested contact)

Ref	Dept / Year	Trend	Contact via	Priority	Status
R-104	CS / III	Sustained high	Phone	HIGH	Open
R-105	Commerce / II	Rising	In person	MEDIUM	Open
R-099	Maths / I	Rising	E-mail	MEDIUM	Contacted

MANAGEMENT INSIGHTS (ANONYMISED)



CHAPTER VIII

CONCLUSION AND FUTURE ENHANCEMENT

8.1 Conclusion

The Student Stress and Wellbeing Early-Warning Assistant provides students with a private, quick way to reflect on their stress every week and receive immediate supportive resources. By combining a standardised questionnaire with sentiment analysis and trend detection, the system identifies sustained or rising stress early, and consent-based referrals ensure that counsellors' limited time goes to students who want help.

The pilot showed good acceptance by students due to the emphasis on privacy and choice. Anonymised dashboards help management notice stressful periods such as internal examinations and plan support. The system demonstrates responsible use of AI as a support tool rather than a diagnostic one.

8.2 Future Enhancement

- Tamil language check-ins and sentiment analysis.
- Chat-based guided self-help exercises.
- Optional integration with sleep and activity data from wearables (with consent).
- Peer-support volunteer network matching.
- Formal evaluation with the college counselling centre.

CHAPTER IX

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